

Progress Report 1

Measuring the Recall Cost of Approximate Nearest Neighbor Search: A Parameter Selection Heuristic for Vector Databases

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1. Project Overview

Nearly every production system that does semantic search, recommendation, or retrieval-augmented generation relies on approximate nearest neighbor (ANN) indexes, because exact search does not scale. Approximate indexes buy speed by returning some wrong neighbors, and they expose tunable parameters that control that tradeoff. In practice those parameters are usually left at library defaults, and because approximate search still returns plausible-looking results when it misses the true neighbors, the loss produces no error and no visible symptom. The main goal of this project is to measure that hidden loss rigorously and then to test whether good parameters can be predicted from cheap, measurable properties of a dataset instead of found through exhaustive search. The two research questions are how far library defaults sit from the achievable recall-latency frontier, and whether dataset properties such as intrinsic dimensionality carry enough signal to guide parameter selection.

2. Work Completed

The measurement infrastructure is built and the first full measurement study has been run end to end. The codebase is 889 lines of Python across six modules and six entry-point scripts, and it produced 657 timed measurement records covering 219 distinct index configurations, from 30 separately built indexes representing 47 minutes of cumulative index construction time. Concretely, the following is done:

2.1 Ground-truth module and harness validation

Exact k-nearest neighbors are computed by exhaustive search over the full base set, since recall is meaningless without exact reference neighbors. Rather than trusting my implementation, I validated it against the ground truth shipped with the ANN-Benchmarks HDF5 files, over all three datasets at their full size (1,000,000 SIFT vectors, 1,183,514 GloVe vectors, 290,000 NYTimes vectors) using 1,000 queries each. This was set up as the gate for trusting every number downstream.

2.2 Index wrappers and parameter sweep

Three index families are wrapped behind one interface so the sweep driver does not need to know which library it is calling: HNSW graphs through `hnswlib`, and IVF-Flat and IVF-PQ through `FAISS`. The driver builds every index in a construction grid once, sweeps the query-

time parameter over each built index, and records tie-aware recall, identifier recall, median and 95th-percentile latency, build time and memory for every configuration, with three timed repeats each. All search is single-threaded and timed per query so that families are comparable on the same machine. The full sweep has been run across all three datasets at 200,000 base vectors with 300 queries at $k = 10$.

2.3 Dataset characterization

The candidate predictive features for the heuristic are implemented and computed: ambient dimensionality, Levina-Bickel maximum-likelihood intrinsic dimensionality, local intrinsic dimensionality measured at the actual query points, and a cluster-structure measure taken from the log-log slope of k-means inertia against cluster count. All four are computable from a 20,000-vector sample in under 30 seconds per dataset, which is the property that would make a heuristic worth having.

2.4 Analysis layer

Pareto frontier extraction, default-versus-frontier comparison, and figure generation are implemented and produce the results in the next section directly from the raw measurement CSVs.

3. Evidence of Progress

3.1 Validation gate: the ground truth is correct

My brute-force neighbors agree with the published ground truth at 99.91 percent on SIFT and 100 percent on GloVe, but only 98.56 percent on NYTimes, which I did not accept at face value. I wrote a diagnostic that recomputed, for every disagreement, the exact distance from the query to my neighbor and to the published one. All 5 SIFT mismatches and all 102 NYTimes mismatches were exact ties, maximum distance gap 0.0. The datasets contain duplicate vectors (387 duplicates in a 20,000-vector NYTimes sample), so two different identifiers are equally correct answers. I added a tie-aware recall definition measured on distances rather than identifiers, under which agreement is 1.0000 everywhere. The gate passes, and the reason it initially appeared not to is now documented rather than hidden.

dataset	identifier agreement@10	tie-aware agreement@10	top-1 mismatches	explained by exact tie
sift-128-euclidean	0.9991	1.0000	5 / 1000	5 / 5
glove-100-angular	1.0000	1.0000	0 / 1000	n/a
nytimes-256-angular	0.9856	1.0000	102 / 1000	102 / 102

Table 1. Harness validation against published ground truth.

3.2 First result: defaults are far from the frontier

Across all nine dataset-by-family combinations, not one library default configuration reached recall 0.75. The mean default recall@10 is 0.455, meaning a developer accepting de-

faults is on average missing more than half of the true nearest neighbors. For the inverted-file indexes the loss is not even a speed tradeoff: at exactly the same latency, a tuned configuration reaches 0.773 instead of 0.528 on SIFT and 0.774 instead of 0.482 on NYTimes, so roughly 25 points of recall are available at zero cost. For HNSW the default sits on its own frontier but at a very low point, and the price of reaching recall 0.95 rises sharply with the dataset: 3.3 times the default latency on SIFT, 17.6 times on GloVe, and 23.2 times on NYTimes. Product quantization at these settings never reached 0.95 on any dataset within the swept grid. Figures 1 and 2 show the full picture.

dataset	family	default recall@10	best recall at same latency	latency for recall 0.95
sift-128-euclidean	HNSW	0.731	0.731	0.144 ms (3.3×)
sift-128-euclidean	IVF-Flat	0.528	0.773	0.334 ms
sift-128-euclidean	IVF-PQ	0.242	0.384	not reached
glove-100-angular	HNSW	0.515	0.515	0.720 ms (17.6×)
glove-100-angular	IVF-Flat	0.501	0.681	1.077 ms
glove-100-angular	IVF-PQ	0.128	0.128	not reached
nytimes-256-angular	HNSW	0.679	0.679	1.792 ms (23.2×)
nytimes-256-angular	IVF-Flat	0.482	0.774	15.100 ms
nytimes-256-angular	IVF-PQ	0.291	0.298	not reached

Table 2. Library defaults against the measured Pareto frontier, $k = 10$, 200,000 base vectors.

3.3 Early signal for the heuristic

The dataset characterization produces a clean ordering. Local intrinsic dimensionality at the query points is 23.9 for SIFT, 43.1 for GloVe and 53.5 for NYTimes, and the HNSW latency required to reach recall 0.95 orders identically at 0.144, 0.720 and 1.792 milliseconds. The k-means inertia slope orders the same way. This is consistent with the hypothesis the project is built on, but three datasets is three points, so I am treating it as a reason to continue rather than as a result.

dataset	ambient dim	intrinsic dim (MLE)	LID at queries	inertia log-slope
sift-128-euclidean	128	19.9	23.9	-0.115
glove-100-angular	100	36.4	43.1	-0.051
nytimes-256-angular	256	46.0	53.5	-0.028

Table 3. Computed dataset properties, the candidate features for the heuristic.

3.4 Measurement stability

Across the 219 aggregated configurations, the relative standard deviation of median query latency over three repeats has a median of 1.3 percent, a mean of 1.9 percent and a maximum of 10.5 percent. That is small relative to the effects being reported, which are multiples rather than percentages.

3.5 Code

The repository contains the harness, the validation scripts, every raw measurement CSV, the characterization output and the figure-generating code, so every number in this report re-generates from the committed data with one command.

```
src/datasets.py src/groundtruth.py src/indexes.py
src/characterize.py src/sweep.py src/frontier.py src/plots.py
scripts/validate_groundtruth.py scripts/investigate_ties.py
scripts/run_all_sweeps.sh scripts/characterize_datasets.py scripts/analyze.py
```

Repository: github.com/chinedu-2002/ann-recall-study

Recall-latency frontiers by index family (200k base vectors, k=10, single thread)

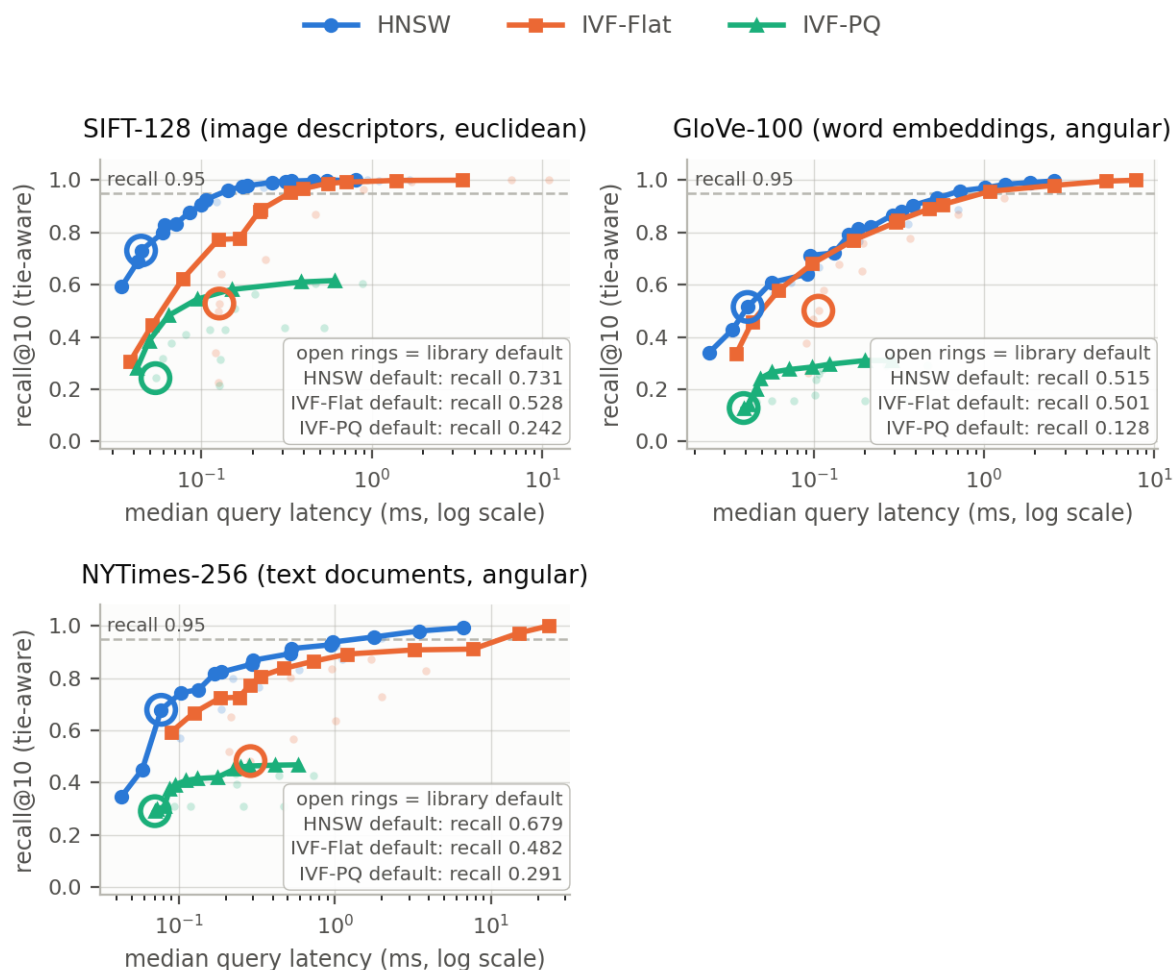


Figure 1. Measured recall-latency frontiers for all three index families on all three datasets. Faint dots are individual measured configurations; solid lines are the Pareto frontier; open rings mark each library's default configuration.

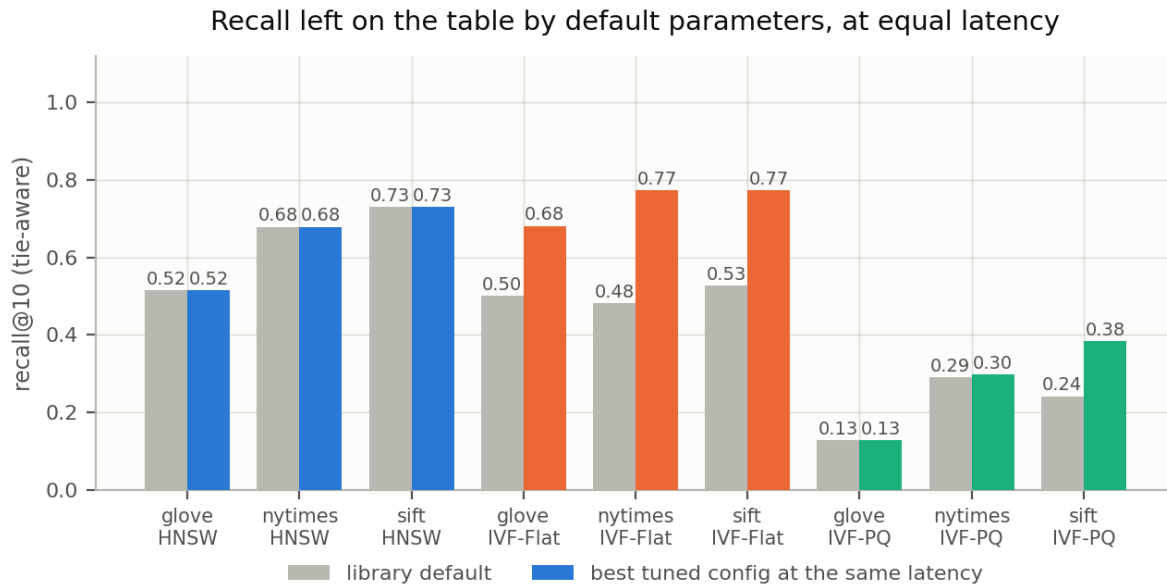


Figure 2. Recall available at no latency cost. Gray is the library default; colored is the best configuration measured at the same or lower latency.

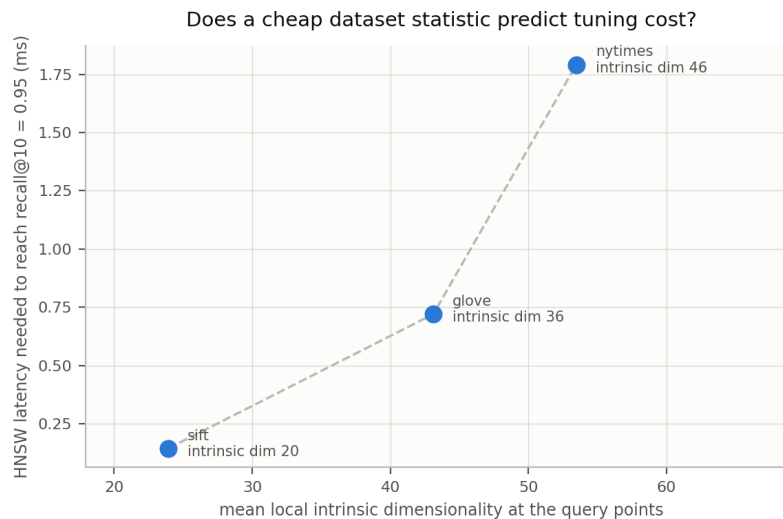


Figure 3. Query-point local intrinsic dimensionality against the HNSW latency needed to reach recall 0.95. Three points, monotone, not yet a fitted model.

4. Challenges

Ground truth that looked wrong but was not. The NYTimes agreement of 98.56 percent could have been dismissed as close enough. Investigating it instead revealed that these corpora contain exact duplicate vectors, which makes identifier-based recall systematically understate a correct result. I addressed it by defining recall on distances instead of identifiers, and I

now report both.

A silent parameter clamp that produced fake flat regions. My first sweep showed identical recall for ef values of 10, 20, 40 and 80, which looked like a real plateau. It was not: hnswlib silently raises ef to at least k, so sweeping ef below k while requesting k neighbors measures the same configuration repeatedly. This also means the documented default ef of 10 is unreachable when retrieving 10 or more neighbors. I fixed the grid to respect the clamp and now record the k actually searched alongside every measurement.

Grids that cannot be shared across datasets. FAISS requires the product-quantization sub-quantizer count to divide the vector dimension exactly, so the grid that worked for 128-dimensional SIFT crashed on 100-dimensional GloVe. I replaced the fixed grid with a target value that snaps down to the nearest divisor of the dimension, and I record the value actually used, since this slightly weakens direct cross-dataset comparison of PQ results.

Estimators that break on duplicate data. The intrinsic dimension estimator returned 7.5 times ten to the eighth on NYTimes. The cause was the same duplicates: a neighbor at distance zero drives the per-point estimate to infinity, and two percent of such points dominate the sample mean. I now drop zero distances, require a minimum count of usable neighbors per point, and aggregate with a trimmed mean, which returns 46.0 and recovers a known intrinsic dimension of 5 on a synthetic control to within 0.3. The cluster-structure measure had a related problem, returning a nonsensical positive slope from MiniBatchKMeans instability on high-dimensional normalized data; full k-means with restarts fixed it.

Compute budget. Single-threaded builds on a two-core machine are slow; the largest single index took 13.8 minutes to construct. I scaled the sweep to 200,000-vector subsets rather than full million-vector collections, which keeps the study tractable at the cost of leaving scale effects for later. Full-size runs are planned for the compute-heavy phase.

5. Next Steps

Before the next progress report I plan to complete four things. First, scale the sweep from 200,000 to the full base sets on at least SIFT and GloVe, to confirm the default-gap finding is not an artifact of subsampling. Second, add $k = 1$ and $k = 100$ passes, since the size of the requested result set interacts with the ef clamp and may change where defaults fall. Third, expand from three datasets to at least six, which is the minimum that makes fitting a heuristic on held-out data meaningful; candidates are Fashion-MNIST, GIST-960, DEEP1B subsets and a sentence-transformer embedding corpus I generate myself, chosen to spread intrinsic dimensionality rather than to add more of the same. Fourth, fit the first version of the heuristic mapping target recall plus dataset features to recommended parameters, and run the held-out evaluation against both library defaults and exhaustive search. I also plan to add a small statistical treatment of the timing variance so that frontier comparisons carry confidence intervals rather than point estimates.